

DAY 3 EXPERIMENTS

Q1.

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori** -N 10 -T 0 -C 0.5 -D 0.05 -U 0.3 -M 0.3 -S -1.0 -Z -c -1

Start Stop

Result list (right-click f...)

08:56:07 - Apriori
08:57:38 - Apriori
08:59:27 - Apriori

Associator output

==== Run information ====

Scheme: weka.associations.Apriori -N 10 -T 0 -C 0.5 -D 0.05 -U 0.3 -M 0.3 -S -1.0 -Z -c -1
Relation: marketbasket
Instances: 10
Attributes: 5
a
b
c
d
e

==== Associator model (full training set) ====

Apriori
=====

Minimum support: 0.3 (3 instances)
Minimum metric <confidences>: 0.5
Number of cycles performed: 14

Generated sets of large itemsets:

Size of set of large itemsets L(1): 5
Size of set of large itemsets L(2): 8
Size of set of large itemsets L(3): 2

Best rules found:

1. a=1 b=1 4 ==> e=1 3 <conf:(0.75)> lift:(0.94) lev:(-0.02) [0] conv:(0.4)
2. c=1 5 ==> a=1 3 <conf:(0.6)> lift:(0.86) lev:(-0.05) [0] conv:(0.5)
3. c=1 5 ==> b=1 3 <conf:(0.6)> lift:(1) lev:(0) [0] conv:(0.67)
4. c=1 5 ==> e=1 3 <conf:(0.6)> lift:(0.75) lev:(-0.1) [-1] conv:(0.33)
5. b=1 e=1 5 ==> a=1 3 <conf:(0.6)> lift:(0.86) lev:(-0.05) [0] conv:(0.5)
6. b=1 6 ==> c=1 3 <conf:(0.5)> lift:(1) lev:(0) [0] conv:(0.75)
7. a=1 e=1 6 ==> b=1 3 <conf:(0.5)> lift:(0.83) lev:(-0.06) [0] conv:(0.6)
8. b=1 6 ==> a=1 e=1 3 <conf:(0.5)> lift:(0.83) lev:(-0.06) [0] conv:(0.6)

Status
OK

Log x 0

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **FPGrowth** -P 2 -I -1 -N 10 -T 0 -C 0.5 -D 0.05 -U 0.3 -M 0.3

Start Stop

Result list (right-click f...)

08:56:07 - Apriori
08:57:38 - Apriori
08:59:27 - Apriori
09:01:59 - FPGrowth

Associator output

==== Run information ====

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.5 -D 0.05 -U 0.3 -M 0.3
Relation: marketbasket
Instances: 10
Attributes: 5
a
b
c
d
e

==== Associator model (full training set) ====

FPGrowth found 8 rules (displaying top 8)

1. [a=1, b=1]: 4 ==> [e=1]: 3 <conf:(0.75)> lift:(0.94) lev:(-0.02) conv:(0.4)
2. [c=1]: 5 ==> [e=1]: 3 <conf:(0.6)> lift:(0.75) lev:(-0.1) conv:(0.33)
3. [c=1]: 5 ==> [a=1]: 3 <conf:(0.6)> lift:(0.86) lev:(-0.05) conv:(0.5)
4. [c=1]: 5 ==> [b=1]: 3 <conf:(0.6)> lift:(1) lev:(0) conv:(0.67)
5. [e=1, b=1]: 5 ==> [a=1]: 3 <conf:(0.6)> lift:(0.86) lev:(-0.05) conv:(0.5)
6. [b=1]: 6 ==> [c=1]: 3 <conf:(0.5)> lift:(1) lev:(0) conv:(0.75)
7. [e=1, a=1]: 6 ==> [b=1]: 3 <conf:(0.5)> lift:(0.83) lev:(-0.06) conv:(0.6)
8. [b=1]: 6 ==> [e=1, a=1]: 3 <conf:(0.5)> lift:(0.83) lev:(-0.06) conv:(0.6)

Status
OK

Log x 0

Q2.

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier
Choose **NaiveBayes**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds 10
☐ Percentage split % 66
More options...

(Nom) buys_computer
Start Stop

Result list (right-click for options)
09:06:18 - bayes.NaiveBayes

Classifier output

```
low      2.0  4.0  
[total]  8.0 12.0  
  
student  
no       5.0  4.0  
yes      2.0  7.0  
[total]  7.0 11.0  
  
credit_rating  
fair     3.0  7.0  
excellent 4.0  4.0  
[total]  7.0 11.0
```

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 8 57.1429 %
Incorrectly Classified Instances 6 42.8571 %
Kappa statistic -0.0244
Mean absolute error 0.4374
Root mean squared error 0.4916
Relative absolute error 91.8631 %
Root relative squared error 99.6492 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.200	0.222	0.333	0.200	0.250	-0.026	0.578	0.557	no
	0.778	0.800	0.636	0.778	0.700	-0.026	0.578	0.697	yes
Weighted Avg.	0.571	0.594	0.528	0.571	0.539	-0.026	0.578	0.647	

=== Confusion Matrix ===
a b <- classified as
1 4 | a = no
2 7 | b = yes

Status
OK

Log x 0

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier
Choose **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds 10
☐ Percentage split % 66
More options...

(Nom) buys_computer
Start Stop

Result list (right-click for options)
09:06:18 - bayes.NaiveBayes
09:06:15 - trees.J48

Classifier output

```
age = <=30  
| student = no: no (3.0)  
| student = yes: yes (2.0)  
age = 31...40: yes (4.0)  
age = >40  
| credit_rating = fair: yes (3.0)  
| credit_rating = excellent: no (2.0)
```

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0.01 seconds

=== Stratified cross-validation ===

Q3

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter Choose **None** Apply Stop

Current relation
Relation: diabetes
Instances: 768
Attributes: 9
Sum of weights: 768

Selected attribute
Name: Pregnancies
Missing: 0 (0%)
Distinct: 17
Type: Numeric
Unique: 2 (0%)

Statistic	Value
Minimum	0
Maximum	17
Mean	3.845
StdDev	3.37

Class: Outcome (Num) Visualize All

Remove

Status OK Log x 0

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier Choose **LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4**

Test options
☐ Use training set
☐ Supplied test set
☒ Cross-validation Folds 10
☐ Percentage split % 66
More options...

(Num) Outcome
Start Stop

Result list (right-click for options)
00:14:00 - functions.LinearRegression

Classifier output
=== Run information ===
Scheme: weka.classifiers.functions.LinearRegression -S 0 -R 1.0E-8 -num-decimal-places 4
Relation: diabetes-weka.filters.unsupervised.attribute.Remove-R1-7
Instances: 768
Attributes: 2
Age
Outcome
Test mode: 10-fold cross-validation
=== Classifier model (full training set) ===
Linear Regression Model
Outcome =
0.0097 * Age +
0.0276
Time taken to build model: 0.04 seconds
=== Cross-validation ===
=== Summary ===
Correlation coefficient 0.2312
Mean absolute error 0.4293
Root mean squared error 0.4637
Relative absolute error 94.3858 %
Root relative squared error 97.2001 %
Total Number of Instances 768

Status OK Log x 0

Q4

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator
Choose **Apriori -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5 -S -1.0 -c -1**

Start Stop

Result list (right-click f...
09:28:32 - Apriori

Associator output

```
M
O
N
K
E
Y
D
A
U
C
I

=== Associator model (full training set) ===

Apriori
=====

Minimum support: 0.85 (4 instances)
Minimum metric <confidence>: 0.8
Number of cycles performed: 3

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 6
Size of set of large itemsets L(3): 1

Best rules found:

1. E=t 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
2. D=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
3. A=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
4. U=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
5. I=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
6. U=f 4 ==> E=t 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
7. E=t 4 ==> U=f 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
8. E=t U=f 4 ==> K=t 4 <conf:(1)> lift:(1) lev:(0) [0] conv:(0)
9. K=t U=f 4 ==> E=t 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
10. K=t E=t 4 ==> U=f 4 <conf:(1)> lift:(1.25) lev:(0.16) [0] conv:(0.8)
```

Status
OK

Log x 0

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator
Choose **FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5**

Start Stop

Result list (right-click f...
09:28:32 - Apriori
09:29:13 - FPGrowth

Associator output

```
=== Run information ===

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.8 -D 0.05 -U 1.0 -M 0.5
Relation: monkey_transactions
Instances: 5
Attributes: 11
M
O
N
K
E
Y
D
A
U
C
I

=== Associator model (full training set) ===

FPGrowth found 7 rules (displaying top 7)

1. [C=f]: 3 ==> [U=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
2. [C=f]: 3 ==> [I=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
3. [N=f]: 3 ==> [D=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
4. [U=f, I=f]: 3 ==> [C=f]: 3 <conf:(1)> lift:(1.67) lev:(0.24) conv:(1.2)
5. [C=f]: 3 ==> [U=f, I=f]: 3 <conf:(1)> lift:(1.67) lev:(0.24) conv:(1.2)
6. [U=f, C=f]: 3 ==> [I=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
7. [I=f, C=f]: 3 ==> [U=f]: 3 <conf:(1)> lift:(1.25) lev:(0.12) conv:(0.6)
```

Status
OK

Log x 0

Q5

Preprocess Classify Cluster Associate Select attributes Visualize

Open file... Open URL... Open DB... Generate... Undo Edit... Save...

Filter
Choose **None** Apply Stop

Current relation
Relation: weather.symbolic
Instances: 14
Attributes: 5
Sum of weights: 14

Attributes
All None Invert Pattern

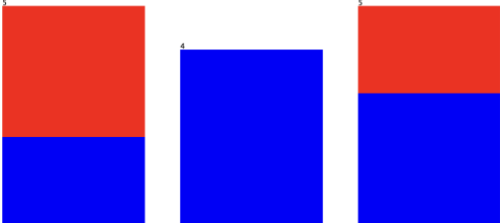
No.	Name
1	outlook
2	temperature
3	humidity
4	windy
5	play

Remove

Selected attribute
Name: outlook
Missing: 0 (0%)
Distinct: 3
Type: Nominal
Unique: 0 (0%)

No.	Label	Count	Weight
1	sunny	5	5
2	overcast	4	4
3	rainy	5	5

Class: play (Nom) Visualize All



Status OK Log x 0

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier
Choose **J48 -C 0.25 -M 2**

Test options
☐ Use training set
☐ Supplied test set Set...
☒ Cross-validation Folds 10
☐ Percentage split % 66
More options...

(Nom) play
Start Stop

Result list (right-click for options)
00:33:27 - trees.J48

Classifier output

```
outlook = sunny
| humidity = high: no (3.0)
| humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)
```

Number of Leaves : 5
Size of the tree : 8
Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===

	Correctly Classified Instances	Incorrectly Classified Instances	Kappa statistic	Mean absolute error	Root mean squared error	Relative absolute error	Root relative squared error	Total Number of Instances
	7	7	-0.0426	0.4167	0.5984	87.5 %	121.2987 %	14

=== Detailed Accuracy By Class ===

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.556	0.600	0.625	0.556	0.588	-0.043	0.633	0.758	yes
	0.400	0.444	0.333	0.400	0.364	-0.043	0.633	0.457	no
Weighted Avg.	0.500	0.544	0.521	0.500	0.508	-0.043	0.633	0.650	

=== Confusion Matrix ===

```
a b ← classified as
5 4 | a = yes
3 2 | b = no
```

Status OK Log x 0

Preprocess Classify Cluster Associate Select attributes Visualize

Classifier

Choose **Logistic -R 1.0E-8 -M -1 -num-decimal-places 4**

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation Folds **10**

☐ Percentage split % **66**

(Nom) play ☐

Result list (right-click for options)

Classifier output

Odds Ratios...

Variable	Class	yes
outlook=sunny		0
outlook=overcast		9.73275593611619E24
outlook=rainy		0.0027
temperature=hot		0.0002
temperature=mild		3.020787521374072E19
temperature=cool		0
humidity=normal		2.0353933107400553E51
windy=FALSE		4.877521304260806E31

Q6

Preprocess Classify Cluster Associate Select attributes Visualize

Associator

Choose **Apriori -N 20 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.33 -S -1.0 -c -1**

Result list (right-click f...)

09:36:35 - Apriori

Associator output

Apriori

=====

Minimum support: 0.55 (3 instances)
Minimum metric <confidence>: 0.6
Number of cycles performed: 9

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6
Size of set of large itemsets L(2): 7
Size of set of large itemsets L(3): 4
Size of set of large itemsets L(4): 1

Best rules found:

1. Chips=t 4 ==> Buns=f 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
2. Buns=f 4 ==> Chips=t 4 <conf:(1)> lift:(1.5) lev:(0.22) [1] conv:(1.33)
3. Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
4. Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
5. Coke=t 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
6. Ketchup=f Coke=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
7. Buns=f Coke=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
8. Buns=f Ketchup=f 3 ==> Coke=t 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
9. Coke=t 3 ==> Buns=f Ketchup=f 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
10. Ketchup=f Chips=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
11. Buns=f Ketchup=f 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
12. Coke=t Chips=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
13. Buns=f Coke=t 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
14. Coke=t 3 ==> Buns=f Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
15. Coke=t Chips=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
16. Ketchup=f Chips=t 3 ==> Coke=t 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
17. Ketchup=f Coke=t 3 ==> Chips=t 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
18. Coke=t 3 ==> Ketchup=f Chips=t 3 <conf:(1)> lift:(2) lev:(0.25) [1] conv:(1.5)
19. Ketchup=f Coke=t Chips=t 3 ==> Buns=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)
20. Buns=f Coke=t Chips=t 3 ==> Ketchup=f 3 <conf:(1)> lift:(1.5) lev:(0.17) [1] conv:(1)

Status

OK  x 0

Preprocess Classify Cluster **Associate** Select attributes Visualize

Associator

Choose **FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.33**

Start Stop

Result list (right-click f...)

09:36:35 - Apriori

09:37:10 - FPGrowth

Associator output

```

=== Run information ===

Scheme: weka.associations.FPGrowth -P 2 -I -1 -N 10 -T 0 -C 0.6 -D 0.05 -U 1.0 -M 0.33
Relation: basket_transactions
Instances: 6
Attributes: 5
  HotDogs
  Buns
  Ketchup
  Coke
  Chips

=== Associator model (full training set) ===

FPGrowth found 5 rules (displaying top 5)

1. [HotDogs=f]: 2 ==> [Buns=f]: 2 <conf:(1)> lift:(1.5) lev:(0.11) conv:(0.67)
2. [Chips=f]: 2 ==> [Coke=f]: 2 <conf:(1)> lift:(2) lev:(0.17) conv:(1)
3. [Ketchup=f]: 4 ==> [Buns=f]: 3 <conf:(0.75)> lift:(1.13) lev:(0.06) conv:(0.67)
4. [Buns=f]: 4 ==> [Ketchup=f]: 3 <conf:(0.75)> lift:(1.13) lev:(0.06) conv:(0.67)
5. [Coke=f]: 3 ==> [Chips=f]: 2 <conf:(0.67)> lift:(2) lev:(0.17) conv:(1)

```

Status

OK

Log x 0

Q7

Preprocess **Classify** Cluster Associate Select attributes Visualize

Classifier

Choose **J48 -C 0.25 -M 2**

Test options

☐ Use training set

☐ Supplied test set Set...

☒ Cross-validation Folds **10**

☐ Percentage split % 66

More options...

(Nom) play

Start Stop

Result list (right-click for options)

00:39:52 - trees J48

Classifier output

```

outlook = sunny
| humidity = high: no (3.0)
| humidity = normal: yes (2.0)
outlook = overcast: yes (4.0)
outlook = rainy
| windy = TRUE: no (2.0)
| windy = FALSE: yes (3.0)

Number of Leaves : 5
Size of the tree : 8

Time taken to build model: 0 seconds

=== Stratified cross-validation ===
=== Summary ===
Correctly Classified Instances 7 50 %
Incorrectly Classified Instances 7 50 %
Kappa statistic -0.0426
Mean absolute error 0.4167
Root mean squared error 0.5984
Relative absolute error 87.5 %
Root relative squared error 121.2987 %
Total Number of Instances 14

=== Detailed Accuracy By Class ===

```

	TP Rate	FP Rate	Precision	Recall	F-Measure	MCC	ROC Area	PRC Area	Class
	0.556	0.600	0.625	0.556	0.588	-0.043	0.633	0.758	yes
	0.400	0.444	0.333	0.400	0.364	-0.043	0.633	0.457	no
Weighted Avg.	0.500	0.544	0.521	0.500	0.508	-0.043	0.633	0.650	

```

=== Confusion Matrix ===
a b <- classified as
5 4 | a = yes
3 2 | b = no

```

Status

OK

Log x 0

Preprocess

Classify

Cluster

Associate

Select attributes

Visualize

Classifier

ChooseJRip -F 3 -N 2.0 -O 2 -S 1

Test options

☐ Use training set

☐ Supplied test set

☒ Cross-validation

☐ Percentage split

Set...

Folds10

%66

More options...

(Nom) play

Start

Stop

Result list (right-click for options)

09:39:52 - trees.J48

09:40:33 - rules.JRip

Classifier output

play

Test mode: 10-fold cross-validation

== Classifier model (full training set) ==

JRIP rules:

=====

(humidity = high) and (outlook = sunny) => play=no (3.0/0.0)

(outlook = rainy) and (windy = TRUE) => play=no (2.0/0.0)

=> play=yes (9.0/0.0)

Number of Rules : 3

Time taken to build model: 0.01 seconds